ENERGY EFFICIENCY MODULE-HEAT EXCHANGERS

HEAT EXCHANGER PERFORMANCE ASSESSMENT

Heat Exchangers can deteriorate over time due to off-design operation, fouling, scaling, or corrosion.

This tool periodically assesses thermal duty, heat transfer coefficients (U), effectiveness (ϵ), and hydraulic pressure drops (ΔP) for both Shell & Tube and Plate Exchangers.

Metadata & Operational Configuration

Select Heat Exchanger Type:

Plate Heat Exchanger (PHE)

Design Data Inputs (from Test/Spec Reports)

Flow Configuration:

Counterflow

No. of Hot Side Passes

1

No. of Cold Side Passes

1

Total Design Heat Duty (kW)

6139.53

Overall Heat Transfer Coefficient U_d (Design/Service) (kW/m^2-K)

7.2350

Design Fouling Margin (%)

Hot Fluid Side Pressure Drop Design (bar)

0.65

Cold Fluid Side Pressure Drop Design (bar)

0.98

Hot Fluid Temp Range (°C)

19.10

Cold Fluid Temp Range (°C)

16.90

Corrected LMTD Design (°C)

5.00

15.0

Operation Data Inputs

Hot Fluid Flow Inlet (kg/hr)
180000.00

Hot Fluid Temp Inlet (°C)
56.10

Hot Fluid Pressure Inlet (bar g)
4.50

Cold Fluid Flow Inlet (kg/hr)
220000.00

Hot Fluid Temp Outlet (°C)
39.00

Hot Fluid Pressure Outlet (bar g)
3.30

Cold Fluid Temp Inlet (°C)
32.00

Cold Fluid Pressure Inlet (bar g)
3.50

Cold Fluid Temp Outlet (°C)
43.00

Cold Fluid Pressure Outlet (bar g)
2.20

Fluid Properties

Hot Fluid Specific Heat Capacity (kJ/kg-°C)
4.18700

Latent Heat of Condensation (kJ/kg)
0.00000

Cold Fluid Specific Heat Capacity (kJ/kg-°C)
4.00000

Latent Heat of Vaporisation (kJ/kg)
0.00000

Heat Transfer Area (A) Calculation

How do you want to provide the Heat Transfer Area (A) m^2?

- Calculate A (Plate Geometry)
- Input A Manually

Enter Effective Plate Surface Area (A in m^2)
170.00

Final Calculated Heat Transfer Area (A): 170.0000 m²

How do you want to provide the Correction Factor (F)?

- (•) Default PHE Factor (F = 0.95)
- ( ) Calculate F (Multi-pass PHE)
- ( ) Input F Manually

Assessment Results

	Parameters	Units	Operation Data	Design Data
0	Heat Exchanger Type	-	Plate Heat Exchanger (PHE)	Plate Heat Exchanger (PHE)
1	Duty (Q)	kW	3579.89	6139.53
2	Hot Fluid Pressure Drop ΔPh	bar	1.20	0.65
3	Cold Fluid Pressure Drop ΔPc	bar	1.30	0.98
4	Temp Range Hot Fluid	°C	17.10	19.10
5	Temp Range Cold Fluid	°C	11.00	16.90
6	Corrected LMTD	°C	9.25	5.00
7	Heat Transfer Coefficient U	kW/m²-K	2.277	7.235

Effectiveness (ε)
0.7095

Capacity Ratio (r)
0.8564

Fouling Resistance (Rd)
0.3009 m²K/kW

AI Performance Diagnostic Summary

Generate AI Assessment Summary

Diagnostic Summary

- **Fouling Margin Assessment:** The operating overall heat-transfer coefficient ($U_{op} = 2.277\ kW\,m^{-2}\,K^{-1}$) is far below the design value ($U_d = 7.235\ kW\,m^{-2}\,K^{-1}$) and the clean-condition value ($U_clean = 8.320\ kW\,m^{-2}\,K^{-1}$). The thermal drop versus clean is 72.6 %, which exceeds the 15 % design fouling allowance; therefore the fouling margin is **exhausted** and performance is well outside the allowable range.
- **Thermal Effectiveness & Capacity Assessment:** The unit is operating at a thermal effectiveness of **E = 0.7095** with a capacity ratio of **r = 0.8564**. Because r is close to unity, the exchanger is in a **balanced-flow regime**, allowing relatively efficient heat transfer, but the effectiveness of ~71 % indicates that a substantial portion of the available temperature driving force is not being utilized—consistent with the degraded U-value.
- **Hydraulic Comparison:** Both hot-side and cold-side pressure drops are elevated: $\Delta P_h = 1.20$ bar (design = 0.65 bar) and $\Delta P_c = 1.30$ bar (design = 0.98 bar). The 85 % increase on the hot side and 33 % increase on the cold side point to significant flow resistance, likely from fouling or plate blockage.
- **Maintenance Recommendation:** Because the fouling margin is exhausted and pressure drops are abnormally high, schedule **chemical cleaning (CIP) of the plate bundle** followed by a **visual plate inspection** for localized scaling, corrosion, or mechanical damage. If cleaning does not restore U-values, consider plate replacement or redesign of flow distribution.

Brief Engineering Insight

The steep decline in U_{op} (≈ 73 % below clean) reduces the effective heat-transfer area, forcing the exchanger to operate with a lower LMTD (9.25 °C vs. design 5.00 °C) to meet the same duty, which in turn depresses thermal effectiveness to ~0.71 despite a near-balanced capacity ratio ($r \approx 0.86$). The calculated fouling resistance ($R_d \approx 0.30\ m^2\,K\,kW^{-1}$) indicates a thick, insulating layer—likely scaling or bio-fouling—that impedes convective heat transfer and raises hydraulic resistance, as evidenced by the elevated ΔP. The combination of high fouling resistance and flow imbalance limits the exchanger's ability to extract the available heat, necessitating cleaning to restore both thermal and hydraulic performance.